

Monetary Policy and Global Uncertainty in Indonesia as a Small Open Economy: ARDL Evidence with Structural Breaks

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Abstract

This study examines how monetary policy in Indonesia responds to rising global uncertainty and structural geopolitical disruptions within the framework of a small open emerging economy. Using annual data from 1985 to 2025, the study employs the Autoregressive Distributed Lag (ARDL) approach incorporating structural break dummies associated with the 1997–1998 Asian Financial Crisis, the 2008 Global Financial Crisis, and the recent geopolitical tensions linked to the Russia–Ukraine conflict and escalating Middle East instability during 2022–2025. Unit root and cointegration tests are conducted to determine the long-run equilibrium relationship among the variables, while the Error Correction Model (ECM) representation of the ARDL framework captures both short-run dynamics and the speed of adjustment toward equilibrium. Granger causality analysis is further applied to identify short-run predictive relationships between monetary policy indicators and macroeconomic variables.

The findings reveal that global uncertainty significantly weakens the effectiveness of monetary policy transmission to economic growth and macroeconomic stability in Indonesia. External geopolitical shocks increase exchange rate volatility, capital flow instability, and inflationary pressures, thereby complicating monetary policy management in an emerging small open economy. Nevertheless, inflation control, exchange rate stabilization, and the accumulation of foreign exchange reserves continue to exert a positive long-run influence on economic resilience. The presence of structural breaks substantially changes the magnitude and direction of monetary policy transmission, indicating asymmetric and uneven adjustment processes across different crisis periods. The results further suggest that Indonesia's monetary system remains highly vulnerable to external geopolitical and financial disturbances, emphasizing the importance of credible, adaptive, and well-coordinated monetary and fiscal policy frameworks to maintain macroeconomic stability amid increasing global uncertainty and geopolitical fragmentation.

Keywords: ARDL; global uncertainty; monetary policy; Indonesia; small open economy; structural breaks

Subjects: Asian Economics; Macroeconomics; Monetary Economics; Econometrics

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1. Introduction

Global economic uncertainty has increased significantly over recent decades due to various global disruptions, including geopolitical conflicts, financial instability, supply chain fragmentation, energy price volatility, and disturbances in international trade and capital markets. These conditions have weakened investment decisions, increased exchange rate pressures, and complicated the transmission mechanism of monetary policy (Baker et al., 2016; Bloom, 2009; Caldara & Iacoviello, 2022). The effects of global uncertainty are particularly severe for small open emerging economies such as Indonesia, where macroeconomic stability remains highly sensitive to capital inflows, exchange rate fluctuations, commodity prices, and external trade performance (Luk et al., 2020). Compared with advanced economies, small open economies generally possess more limited policy flexibility and are more vulnerable to external financial shocks. Consequently, the effectiveness of monetary policy is strongly influenced by external conditions, structural resilience, and institutional credibility.

In recent years, geopolitical tensions such as the Russia–Ukraine conflict, the escalation of instability in the Middle East, and increasing fragmentation in the global trading system have intensified inflationary pressures and exchange rate volatility across emerging economies. Structural disruptions of this kind frequently generate regime shifts that alter the relationship between inflation, exchange rates, foreign reserves, and economic growth (Gupta, 2025; Xu & Xu, 2022). Empirical evidence from several ASEAN and emerging economies indicates that global policy uncertainty weakens conventional monetary policy transmission channels and reduces macroeconomic effectiveness during periods of crisis and external turbulence (Eklou, 2023).

Indonesia represents an important case of a small open emerging economy that is highly exposed to external shocks and global financial uncertainty. As one of the largest economies in Southeast Asia, Indonesia relies heavily on international trade, commodity exports, portfolio capital inflows, and exchange rate stability. Historical episodes such as the 1997–1998 Asian Financial Crisis and the 2008 Global Financial Crisis demonstrated how external disturbances can rapidly weaken domestic financial stability, increase inflationary pressures, and reduce economic growth. More recently, geopolitical tensions during 2022–2025, including disruptions in global energy markets and supply chains resulting from the Russia–Ukraine war and Middle East conflicts, have increased imported inflation risks and exchange rate volatility in Indonesia.

Although Bank Indonesia has actively implemented inflation-targeting policies, exchange rate stabilization measures, and foreign reserve accumulation strategies, the effectiveness of monetary policy transmission remains challenged under conditions of heightened global uncertainty. External shocks have frequently triggered capital outflows, rupiah depreciation pressures, and fluctuations in commodity prices, complicating macroeconomic management and policy coordination. Recent assessments by international institutions also suggest that emerging economies, including Indonesia, remain vulnerable to sudden global financial tightening and geopolitical fragmentation despite relatively strong macroeconomic fundamentals.

Existing studies in Indonesia generally examine monetary policy, inflation, exchange rates, or economic growth separately. While several studies employ cointegration or ARDL approaches to

analyze monetary transmission mechanisms, only limited research incorporates long historical data combined with global uncertainty indicators and structural break modelling. Most previous studies do not explicitly integrate measures such as the World Uncertainty Index, Geopolitical Risk Index, or structural break dummies associated with major external shocks including the Asian Financial Crisis, the Global Financial Crisis, and recent geopolitical conflicts. Without accounting for structural regime changes, empirical findings may become biased because shifts in economic regimes can alter lag structures, adjustment speeds, and even the direction of relationships among variables.

Furthermore, previous Indonesian studies often treat external shocks as exogenous events without examining how monetary policy effectiveness changes across different uncertainty regimes. There is still limited evidence regarding how global uncertainty interacts with monetary variables under structural breaks, particularly in the context of exchange rate pressures, foreign reserves, inflation dynamics, and economic growth. Likewise, empirical evidence concerning the speed of adjustment toward equilibrium under periods of heightened geopolitical uncertainty remains relatively scarce. Therefore, this study aims to examine the relationship between global uncertainty and monetary policy effectiveness in Indonesia by incorporating structural breaks associated with the 1997–1998 Asian Financial Crisis, the 2008 Global Financial Crisis, and recent geopolitical tensions during 2022–2025 within an ARDL and Error Correction Model (ECM) framework. The study also applies interaction analysis and Granger causality testing to evaluate both short-run and long-run dynamics across different structural regimes.

This study provides several theoretical and policy contributions. From a theoretical perspective, the integration of global uncertainty indicators with structural break modelling within the ARDL framework contributes to the literature on monetary policy transmission in small open emerging economies. Indonesia offers a unique case due to its high exposure to commodity price fluctuations, exchange rate volatility, global capital mobility, and geopolitical spillovers. The findings are expected to enrich comparative studies among ASEAN and emerging market economies regarding the vulnerability of monetary systems under global uncertainty. From a policy perspective, this study provides empirical evidence that can assist policymakers and monetary authorities in designing more adaptive, credible, and resilience-oriented monetary policy frameworks. Methodologically, the incorporation of structural break dummies, interaction effects, and uncertainty measures into the ARDL framework offers a more comprehensive approach for analyzing monetary policy effectiveness under changing global economic regimes.

2. Review

In this review section, the study employs theoretical, empirical, and policy-oriented perspectives to establish the analytical foundation for modeling choices, interpretation of findings, and monetary policy analysis in the Indonesian context.

2.1. Theoretical Review

The classical small open economy framework developed by Robert Mundell and Marcus Fleming explains the fundamental trade-off between exchange rate stability, monetary policy independence, and capital mobility. According to the Mundell–Fleming model, a country cannot simultaneously achieve a fixed exchange rate regime, free capital mobility, and fully independent monetary policy, commonly referred to as the “impossible trinity” or policy trilemma (Fleming, 1962; Mundell, 1960).

For emerging economies such as Indonesia, this framework remains highly relevant because the domestic monetary system is deeply integrated with global financial markets and highly sensitive to external capital flows and exchange rate volatility.

Although Indonesia formally adopts a managed floating exchange rate regime and an inflation-targeting framework under Bank Indonesia, monetary policy autonomy remains constrained by global financial conditions, especially during periods of heightened uncertainty and capital market turbulence. Under conditions of increasing global uncertainty, external shocks frequently trigger exchange rate depreciation, portfolio capital reversals, and imported inflation pressures, thereby reducing the effectiveness of domestic monetary policy transmission. Consequently, structural disturbances originating from geopolitical conflicts, global financial tightening, commodity price volatility, or international trade disruptions can substantially weaken the ability of monetary authorities to stabilize inflation and economic growth.

Modern Dynamic Stochastic General Equilibrium (DSGE) models for small open economies further extend the Mundell-Fleming framework by incorporating uncertainty shocks, exchange rate dynamics, and expectations into monetary policy analysis. Studies such as Justiniano and Preston (2010) demonstrate that optimal monetary policy responses in small open economies may become less aggressive toward exchange rate fluctuations when uncertainty and parameter instability increase, particularly under local currency pricing mechanisms. These theoretical developments provide important support for incorporating structural breaks and interaction effects into empirical models because the transmission mechanism of monetary policy often changes across different economic regimes.

Similarly, exchange rate pass-through theory suggests that fluctuations in exchange rates do not always translate immediately into domestic inflation, especially when producers and exporters adjust prices in local currency terms rather than producer currency pricing (Devereux & Engel, 2003). In the Indonesian context, this phenomenon is particularly relevant because imported inflation dynamics depend heavily on global commodity prices, energy imports, and exchange rate stability. During periods of geopolitical uncertainty and global supply chain fragmentation, the rupiah exchange rate often experiences volatility, yet the transmission to domestic inflation may vary depending on market structure, government intervention, and inflation expectations.

Furthermore, empirical studies in small open emerging economies reveal that domestic macroeconomic variables frequently exhibit asymmetric and inconsistent responses to external shocks and exchange rate disturbances, even under relatively flexible exchange rate systems (Justiniano & Preston, 2010). This theoretical perspective helps explain why monetary policy effectiveness in Indonesia may differ across crisis periods, particularly regarding the interaction between interest rates, exchange rates, inflation, foreign reserves, and economic growth. Structural disruptions such as the 1997–1998 Asian Financial Crisis, the 2008 Global Financial Crisis, and recent geopolitical tensions associated with the Russia-Ukraine conflict and Middle East instability have significantly altered the monetary transmission mechanism in Indonesia.

Another important theoretical perspective emphasizes that the effectiveness of monetary policy is not constant over time but depends heavily on uncertainty regimes and economic conditions. During periods of heightened uncertainty, conventional monetary policy instruments may become less effective, requiring more adaptive and unconventional policy responses. Research in the European

monetary area indicates that unconventional monetary policies tend to perform better during periods of elevated uncertainty and financial instability (Hauzenberger et al., 2021). Likewise, Trabelsi (2025) demonstrates that global uncertainty significantly affects monetary policy communication and policy effectiveness through time-varying transmission channels.

These theoretical arguments provide strong justification for the use of structural break dummies, interaction terms, and uncertainty indices in empirical modeling. In the Indonesian context, incorporating structural breaks associated with the Asian Financial Crisis, the Global Financial Crisis, and recent geopolitical tensions allows the analysis to capture regime-specific changes in monetary policy effectiveness. Therefore, the application of the ARDL framework combined with structural break analysis and interaction effects becomes particularly relevant for understanding how global uncertainty reshapes monetary policy transmission in Indonesia as a small open emerging economy.

2.2. Expected Signs and Transmission Channels of Global Uncertainty

Drawing upon the theoretical frameworks discussed above, this study expects global uncertainty to affect Indonesia's macroeconomy through several major transmission channels with identifiable directional effects. As a small open emerging economy, Indonesia remains highly integrated with global trade, commodity markets, capital flows, and international financial systems. Consequently, rising global uncertainty is expected to influence economic growth, monetary policy effectiveness, exchange rate stability, inflation dynamics, and foreign exchange reserves.

First, higher global uncertainty is expected to reduce Indonesia's real economic growth. Periods of heightened uncertainty generally cause firms, investors, and households to delay investment, production expansion, and consumption decisions because future economic conditions become more difficult to predict (Bloom, 2009; Gupta, 2025). In Indonesia, this "wait-and-see" behavior becomes more pronounced because economic growth is closely linked to export performance, foreign portfolio investment, commodity prices, and global trade demand. External shocks such as geopolitical conflicts, global financial tightening, and disruptions in energy and commodity markets tend to weaken investment confidence and reduce aggregate demand. Therefore, the expected a priori sign of the World Uncertainty Index (WUIIND) or Geopolitical Risk Index is negative toward real GDP growth (Hernandez et al., 2024; Luk et al., 2020).

Second, rising global uncertainty is expected to weaken the monetary policy transmission mechanism in Indonesia. Within the Mundell–Fleming framework, countries with relatively open capital markets and managed floating exchange rate systems often face limitations in maintaining full monetary autonomy under conditions of volatile global capital mobility (Mundell, 1960). Although Indonesia formally adopts an inflation-targeting regime with exchange rate flexibility, external uncertainty frequently intensifies portfolio capital outflows, exchange rate depreciation, and financial market volatility. During periods of global turbulence, Bank Indonesia often relies on foreign exchange reserves and monetary stabilization measures to reduce excessive rupiah volatility and maintain investor confidence.

Under normal economic conditions, higher foreign exchange reserves are generally expected to support economic stability and growth because they strengthen market confidence, reduce external vulnerability, and improve macroeconomic resilience. Therefore, the long-run coefficient of foreign exchange reserves (LN_FOREIGN_EXCHANGE_RESERVE) is expected to be positive.

However, during periods of elevated uncertainty and geopolitical instability, reserves may function more defensively to stabilize the exchange rate and mitigate capital flight pressures. Consequently, the short-run interaction between foreign reserves and uncertainty is expected to generate weaker or potentially negative effects on growth and monetary transmission.

Third, global uncertainty also operates through the inflation and cost transmission channel. Geopolitical conflicts and supply chain disruptions frequently increase global commodity and energy prices, thereby generating imported inflation pressures in emerging economies (Baker et al., 2016; Caldara & Iacoviello, 2022). In Indonesia, fluctuations in oil prices, food prices, shipping costs, and exchange rates significantly influence domestic inflation dynamics because of the economy's dependence on imported energy, raw materials, and global supply chains. In the short run, rising inflation is expected to reduce purchasing power, suppress consumption, and weaken economic growth, indicating a negative sign for lagged changes in the Consumer Price Index (CPI). Nevertheless, in the long run, moderate and stable inflation may still be associated with financial deepening, rising nominal incomes, and expanding economic activity, consistent with previous ARDL findings in emerging economies regarding the positive long-run relationship between inflation and output under stable macroeconomic conditions.

Fourth, these transmission channels are expected to be highly regime-dependent. Major structural shocks such as the 1997–1998 Asian Financial Crisis, the 2008 Global Financial Crisis, and recent geopolitical tensions during 2022–2025 associated with the Russia–Ukraine conflict and instability in the Middle East are likely to intensify the negative impact of uncertainty on economic growth and alter the effectiveness of monetary policy transmission. Such shocks may change the role of foreign reserves, exchange rate stabilization mechanisms, inflation dynamics, and the speed of adjustment toward long-run equilibrium. Therefore, the inclusion of structural break dummies and interaction terms is theoretically justified because the relationships among macroeconomic variables are expected to vary across different crisis regimes (Gupta, 2025; Hernandez et al., 2024; Justiniano & Preston, 2010).

Taken together, these theoretical perspectives—including the policy trilemma framework, DSGE uncertainty modeling, exchange rate pass-through theory, and nonlinear regime-dependent monetary transmission—provide a strong conceptual foundation for interpreting the empirical findings of this study. These approaches are particularly relevant for understanding why monetary policy effectiveness in Indonesia may weaken during periods of heightened global uncertainty, structural disruptions, and geopolitical instability.

2.3. Empirical Review

The relationship between global uncertainty and monetary policy effectiveness has received increasing empirical attention in recent years. Globally, Hernandez et al. (2024) argue that rising global uncertainty significantly reduces the effectiveness of monetary policy in emerging and developing economies, particularly during periods characterized by supply chain disruptions, geopolitical conflicts, and financial market instability. Similarly, Lakdawala et al. (2021) demonstrate that uncertainty amplifies the spillover effects of United States monetary policy toward emerging economies through exchange rate pressures, capital flow volatility, and bond yield adjustments.

In the context of emerging Asian economies, empirical evidence increasingly suggests that global uncertainty negatively affects economic growth and complicates macroeconomic stabilization efforts. Studies employing ARDL and panel-data approaches indicate that uncertainty-driven shocks weaken investment activity, disrupt trade flows, and reduce the effectiveness of conventional monetary policy instruments. Nevertheless, economies with stronger institutional credibility, deeper financial markets, and more resilient external sectors tend to experience lower macroeconomic vulnerability under global shocks.

For Indonesia, a growing number of studies have examined monetary policy transmission, exchange rate dynamics, inflation targeting, and macroeconomic stability. Several studies employing ARDL, Vector Error Correction Models (VECM), and cointegration techniques find that monetary variables such as interest rates, money supply, inflation, and exchange rates exhibit significant long-run relationships with economic growth and external balance stability. Other empirical studies also highlight the importance of foreign reserves and exchange rate management in maintaining macroeconomic resilience under periods of external volatility.

However, despite the expanding literature, several important gaps remain. First, most Indonesian studies analyze monetary variables, inflation, exchange rates, or economic growth separately without explicitly incorporating global uncertainty indicators such as the World Uncertainty Index or Geopolitical Risk Index. Second, only limited empirical research integrates structural break analysis associated with major external crises such as the 1997–1998 Asian Financial Crisis, the 2008 Global Financial Crisis, and recent geopolitical tensions during 2022–2025. Ignoring structural regime shifts may lead to biased estimation results because monetary transmission mechanisms often change significantly across different crisis periods.

Third, previous Indonesian studies rarely examine interaction effects between global uncertainty and monetary policy variables within a unified ARDL framework. Most studies treat external shocks merely as exogenous events without testing how the effectiveness of monetary policy changes across uncertainty regimes. Likewise, limited evidence exists regarding the speed of adjustment toward long-run equilibrium under conditions of geopolitical instability and global financial uncertainty. Granger causality analysis across structural break regimes also remains relatively underexplored in the Indonesian context.

Therefore, this study contributes to the literature by incorporating global uncertainty measures, structural break dummies associated with the Asian Financial Crisis, the Global Financial Crisis, and recent geopolitical tensions, as well as interaction effects within an ARDL-ECM framework. By doing so, the study aims to provide a more comprehensive understanding of how global uncertainty reshapes monetary policy transmission and macroeconomic stability in Indonesia as a small open emerging economy.

Within the framework of a small open economy, the policy review section further examines Indonesia's monetary policy framework and the capacity of Bank Indonesia to respond adaptively to increasing global uncertainty and geopolitical disruptions, as presented in Table 1

Table 1. Policy Review of Indonesia's Monetary Policy and External Stability in Relation to Global Uncertainty

Time/Period	Key Policy Features/Instruments	Strengths and Relevance to Indonesia as a Small Open Emerging Economy	Gaps and Challenges under Global Uncertainty
Bank Indonesia Monetary Policy Framework	Bank Indonesia implements an inflation-targeting framework supported by interest rate policy (BI-Rate/BI7DRR), open market operations, macroprudential instruments, reserve requirements (GWM), and foreign exchange stabilization measures.	The framework supports price stability, exchange rate management, and macroeconomic resilience while maintaining credibility in financial markets. It is relatively flexible compared to fixed exchange rate systems.	Monetary autonomy remains constrained by global capital flows, exchange rate volatility, and external financial tightening. Policy effectiveness may weaken during periods of geopolitical fragmentation and global uncertainty.
Recent Monetary Policy Stance (2022–2025)	Bank Indonesia adopted a tighter monetary stance through policy rate adjustments, exchange rate stabilization, and liquidity management to respond to global inflation, Fed tightening, and geopolitical risks linked to the Russia-Ukraine conflict and Middle East tensions.	The adaptive and forward-looking policy stance strengthens investor confidence, stabilizes inflation expectations, and protects external balances amid volatile global conditions.	Persistent global uncertainty and commodity price volatility may reduce the effectiveness of interest rate transmission and increase trade-offs between growth stabilization and inflation control.
Monetary Transmission and Operating Target	Bank Indonesia utilizes interest rate channels, liquidity absorption, exchange rate interventions, and macroprudential instruments to influence inflation, credit growth, and financial stability.	Multiple transmission channels provide flexibility in responding to domestic and external shocks, especially in maintaining rupiah stability and controlling imported inflation.	Transmission mechanisms may become weaker during periods of high uncertainty because of delayed banking sector responses, volatile capital flows, and fragmented global financial conditions.
Bank Lending and Credit Channel Policies	Policies include reserve requirement adjustments, macroprudential liquidity incentives, loan-to-value (LTV) regulation, and banking sector liquidity support to maintain credit expansion and financial system stability.	These measures support banking sector resilience and sustain credit distribution during periods of economic slowdown or financial stress.	Commercial banks may become more risk-averse during uncertainty periods, weakening credit transmission and reducing the effectiveness of expansionary monetary measures.
Foreign Exchange Policy and Reserve Management	Bank Indonesia actively intervenes in the foreign exchange market, manages foreign exchange reserves, and conducts Domestic Non-Deliverable Forward (DNDF) operations to stabilize the rupiah exchange rate.	Foreign exchange reserves serve as an important buffer against external shocks, capital outflows, and exchange rate volatility in a globally integrated economy.	Under prolonged geopolitical tensions and global financial instability, reserve buffers may face significant pressure, while excessive intervention may reduce policy flexibility and increase external vulnerability.
Coordination between Monetary and Fiscal Policy	Monetary authorities coordinate with the Government of Indonesia through inflation control teams, fiscal stimulus alignment, and financial system stabilization policies.	Strong policy coordination enhances macroeconomic resilience, supports investor confidence, and improves crisis management capacity during external shocks.	Policy coordination challenges may emerge when global uncertainty simultaneously pressures inflation, fiscal deficits, exchange rates, and economic growth, creating policy trade-offs.
Digital Finance and Financial System Stability	Bank Indonesia promotes payment system digitalization, QRIS expansion, local currency settlement cooperation, and financial market deepening to strengthen monetary transmission and economic resilience.	Digital financial integration increases transaction efficiency, financial inclusion, and domestic payment resilience against external financial disruptions.	Rapid digital financial expansion may also increase exposure to cyber risks, capital mobility volatility, and financial contagion during periods of global uncertainty.

Sources: Bank Indonesia (2022, 2023, 2024, 2025); IMF Reports on Indonesia; World Bank (2023, 2024); Asian Development Bank (2024); Caldara & Iacoviello (2022); Baker et al. (2016)

The literature on central bank independence in Indonesia suggests that the autonomy and credibility of Bank Indonesia play a crucial role in maintaining monetary stability, anchoring inflation expectations, and protecting monetary policy from excessive political and fiscal pressures. Since the implementation of the inflation-targeting framework and institutional reforms following the 1997–1998 Asian Financial Crisis, Bank Indonesia has gradually strengthened its credibility and operational independence in conducting monetary policy. Greater institutional autonomy is widely considered essential for improving policy effectiveness, strengthening market confidence, and ensuring macroeconomic stability in periods of economic uncertainty and financial turbulence.

Nevertheless, despite significant institutional progress, monetary policy effectiveness in Indonesia continues to face several structural constraints. External vulnerabilities, dependence on global capital flows, exchange rate pressures, commodity price volatility, and global financial tightening frequently limit the ability of monetary authorities to respond optimally during crisis periods. Political-economic pressures, fiscal coordination challenges, and market sensitivity toward global monetary conditions also affect the degree of effective policy independence, particularly under high-uncertainty regimes. This perspective aligns with the findings of this study, which indicate that monetary policy effectiveness tends to weaken during periods of heightened global uncertainty and geopolitical instability, especially when exchange rate volatility and external financial shocks intensify.

Recent studies on Indonesia’s monetary framework further emphasize the trade-off between macroeconomic stability, financial inclusion, and economic growth. Expansionary and accommodative monetary policies are often designed to support credit expansion, investment, and economic recovery. However, under conditions of elevated global uncertainty, monetary authorities frequently prioritize exchange rate stability, inflation control, and financial market confidence over aggressive growth-oriented policies. Consequently, periods of severe global uncertainty and geopolitical disruption tend to increase the trade-off between economic stabilization and growth support. This condition may alter the responsiveness of monetary policy instruments and influence the magnitude and direction of estimated coefficients within empirical models.

The policy review therefore highlights that, despite continuous improvements in Indonesia’s monetary policy framework, external shocks, structural vulnerabilities, and uncertainty originating from global financial and geopolitical developments continue to constrain the effectiveness of monetary transmission. In particular, the 1997–1998 Asian Financial Crisis, the 2008 Global Financial Crisis, and recent geopolitical tensions during 2022–2025 demonstrate that external disturbances can significantly reshape inflation dynamics, exchange rate stability, capital flows, and economic growth in Indonesia. Accordingly, this study seeks to provide an empirical evaluation of Indonesia’s monetary policy effectiveness by situating the country’s monetary system within the broader dynamics of global uncertainty, structural breaks, and regime-dependent monetary transmission.

Based on the theoretical framework and empirical gaps identified in the Indonesian context, the following hypotheses are formulated to investigate the interaction between monetary policy, global uncertainty, and structural breaks:

Hypotheses

H1: After controlling for inflation and foreign exchange reserves, global uncertainty exerts a negative long-run effect on Indonesia's economic growth.

H2: The effectiveness of monetary policy transmission through foreign exchange reserves, liquidity, and exchange rate stabilization weakens during periods of elevated global uncertainty (the interaction term between monetary policy variables and uncertainty is significantly negative).

H3: The impact of monetary policy and its interaction with global uncertainty differs across structural break regimes, particularly during the post-1998 Asian Financial Crisis, post-2008 Global Financial Crisis, and post-2022 geopolitical uncertainty periods.

H4: The speed of adjustment toward long-run equilibrium becomes slower during periods characterized by high global uncertainty and structural disruptions.

Accordingly, the following methodology section outlines the ARDL and Error Correction Model (ECM) approach designed to test these hypotheses, with particular emphasis on structural breaks, interaction effects, and regime shifts affecting monetary policy transmission in Indonesia as a small open emerging economy.

3. Methodology

This study employs the Autoregressive Distributed Lag (ARDL) framework to investigate the interaction between monetary policy and global uncertainty in Indonesia as a small open emerging economy. The ARDL approach is selected because of its efficiency in small samples and its ability to estimate both short-run and long-run relationships among variables integrated at different orders, specifically $I(0)$ and $I(1)$ (Menegaki, 2019; Nkoro & Uko, 2016; Pesaran et al., 2001). In this study, lag lengths for each variable are automatically determined using the Akaike Information Criterion (AIC). The maximum lag length is restricted to four for annual data in order to preserve degrees of freedom while still capturing the delayed effects of monetary and macroeconomic shocks over multiple periods.

Annual time-series data covering the period from 1985 to 2025 are utilized in this study. The selected time horizon enables the analysis to capture several major structural and regime shifts in Indonesia's macroeconomic history, including the 1997–1998 Asian Financial Crisis, the 2008 Global Financial Crisis, and recent geopolitical disruptions associated with the Russia–Ukraine conflict and Middle East instability during 2022–2025. These events are particularly relevant for identifying structural breaks and regime-dependent monetary policy dynamics.

The empirical model includes Gross Domestic Product (GDP) growth as the dependent variable, representing overall macroeconomic performance and economic stability targeted by monetary policy. Since monetary transmission in a small open economy generally operates through domestic price channels, exchange rate stability, and external balance mechanisms, several explanatory variables are incorporated into the model. These include the Consumer Price Index (CPI), foreign exchange reserves, and a global uncertainty indicator.

Foreign exchange reserves (LN_FOREIGN_EXCHANGE_RESERVE or LN_FXE) are measured using Indonesia's gross official reserve assets held by Bank Indonesia. The variable is transformed into natural logarithmic form to reduce skewness and allow elasticity interpretation of coefficients. Foreign exchange reserves are included because they represent an important monetary stabilization instrument in managing exchange rate volatility, capital flow pressures, and external sector resilience in Indonesia's open economy framework.

The Consumer Price Index (CPI) is included as a proxy for inflation and domestic price dynamics. Inflation plays an essential role in the monetary transmission mechanism because it reflects the interaction between aggregate demand, imported inflation, exchange rate pass-through, and monetary policy responses. In the Indonesian context, CPI dynamics are particularly sensitive to energy prices, food prices, exchange rate fluctuations, and global commodity shocks.

To capture the external uncertainty environment, this study incorporates the World Uncertainty Index for Indonesia (WUIIND) or a comparable global uncertainty indicator obtained from the World Uncertainty Index database and the Federal Reserve Economic Data (FRED) platform. The uncertainty index measures the frequency of uncertainty-related terms in country reports and serves as a proxy for global risk conditions affecting Indonesia's macroeconomic and financial stability (Baker et al., 2016; Hernandez et al., 2024). The inclusion of this variable allows the analysis to evaluate how external uncertainty influences monetary transmission, inflation dynamics, exchange rate stability, and economic growth.

Dummy variables are further incorporated to account for structural breaks associated with major external and domestic shocks. Specifically, structural break dummies are assigned for the 1997–1998 Asian Financial Crisis, the 2008 Global Financial Crisis, and the geopolitical uncertainty period during 2022–2025. These dummy variables are used to capture regime shifts that may alter the relationships among monetary variables, inflation, reserves, exchange rates, and GDP growth. In addition, interaction terms between structural break dummies and core regressors are introduced to examine whether the effects of monetary variables on economic growth differ across different crisis regimes.

Within the context of Indonesia as a small open emerging economy, monetary policy transmission is highly influenced by external financial conditions, capital mobility, exchange rate volatility, and global uncertainty. Following the theoretical foundations of DSGE uncertainty models and exchange rate pass-through theory, policy effectiveness is expected to vary across different economic regimes and periods of heightened uncertainty. Therefore, the ARDL framework combined with structural break analysis and interaction effects is considered appropriate for examining regime-dependent monetary transmission dynamics in Indonesia.

Prior to estimation, the stationarity properties of all variables are examined using the Augmented Dickey–Fuller (ADF) unit root test to ensure that variables are integrated at either $I(0)$ or $I(1)$, consistent with ARDL requirements (Dickey & Fuller, 1979; Pesaran et al., 2001). Structural breaks are identified using the Chow Breakpoint Test and CUSUM of Squares stability tests, which help determine significant regime shifts associated with the Asian Financial Crisis, the Global Financial Crisis, and recent geopolitical disruptions (Chow, 1960; Turner, 2010). The identified break periods are subsequently incorporated into the model through dummy variables and interaction terms.

Several diagnostic and stability tests are conducted to ensure model robustness and specification validity, including serial correlation, heteroskedasticity, normality, and stability tests. Furthermore, the Error Correction Model (ECM) representation of the ARDL framework is employed to estimate short-run dynamics and the speed of adjustment toward long-run equilibrium. A negative and statistically significant Error Correction Term (ECT) coefficient indicates the existence of long-run equilibrium adjustment mechanisms. Granger causality analysis is also performed to identify predictive causal relationships among monetary policy variables, inflation, foreign reserves, uncertainty, and economic growth across different structural regimes.

The selected structural break periods are exogenously justified because they correspond to well-documented macroeconomic and geopolitical disturbances affecting Indonesia's economy. The 1997–1998 Asian Financial Crisis caused severe currency depreciation, inflation escalation, banking sector instability, and economic contraction. The 2008 Global Financial Crisis generated external trade and financial market shocks, while geopolitical tensions during 2022–2025 significantly increased commodity price volatility, exchange rate instability, and external uncertainty pressures. Since the ARDL model satisfies all stability and specification diagnostics after incorporating these structural break dummies, additional endogenous break estimation procedures such as Perron, Zivot-Andrews, or Bai-Perron tests are not considered methodologically necessary.

4. Results and Analysis

The empirical results presented in Table 2 and Figure 1 indicate the presence of significant structural breaks associated with the 1997–1998 Asian Financial Crisis, the 2008 Global Financial Crisis, and the geopolitical uncertainty period during 2022–2025. The unit root test results confirm a mixed order of integration among variables, thereby justifying the application of the ARDL approach.

The cointegration analysis further confirms the existence of a long-run equilibrium relationship among GDP growth, inflation, foreign exchange reserves, and global uncertainty in Indonesia. The findings indicate that Indonesia's economic growth is significantly influenced by inflation dynamics, exchange rate stabilization, reserve accumulation, and global uncertainty conditions. The estimated structural break dummies capture substantial economic disruptions during crisis periods, highlighting Indonesia's vulnerability to external financial shocks, geopolitical instability, and regime-dependent monetary policy transmission.

The results also demonstrate that rising global uncertainty weakens the effectiveness of monetary policy transmission, particularly through exchange rate volatility, capital flow instability, and inflationary pressures. Interaction terms between uncertainty indicators and monetary variables suggest that the impact of foreign reserves and inflation on growth changes significantly during crisis regimes. These findings support the argument that monetary policy effectiveness in Indonesia is not constant over time but highly dependent on external uncertainty conditions and structural economic shifts.

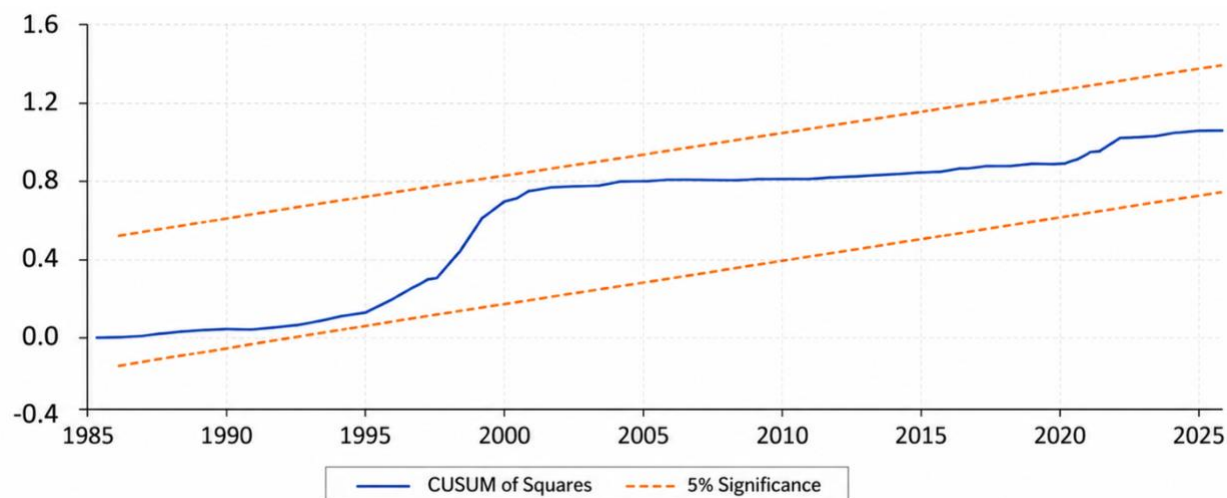
Figure 1. *CUSUM of Squares Tests before Implementing Structural Break Dummies in Indonesia (1985–2025).*

Note. Estimated using E-Views 12. GDP: GROSS_DOMESTIC_PRODUCT; LN_FXE: LN_FOREIGN_EXCHANGE_RESERVE; CPI: CONSUMER_PRICE_INDEX; WUIIND: WORLD_UNCERTAINTY_INDEX_FOR_INDONESIA.

D1998: DUMMY_1997_1998 (Asian Financial Crisis);

D2008: DUMMY_2008 (Global Financial Crisis);

D2022: DUMMY_2022 (Geopolitical Uncertainty and Global Supply Chain Disruption).



The CUSUM of Squares test indicates the presence of structural instability and regime shifts in Indonesia's monetary transmission mechanism, particularly during the Asian Financial Crisis, the Global Financial Crisis, and the recent geopolitical uncertainty period. The results justify the inclusion of structural break dummy variables within the ARDL framework to capture changes in macroeconomic relationships across different crisis regimes.

Table 2. Structural Break Test Results for Indonesia's Monetary Policy Model

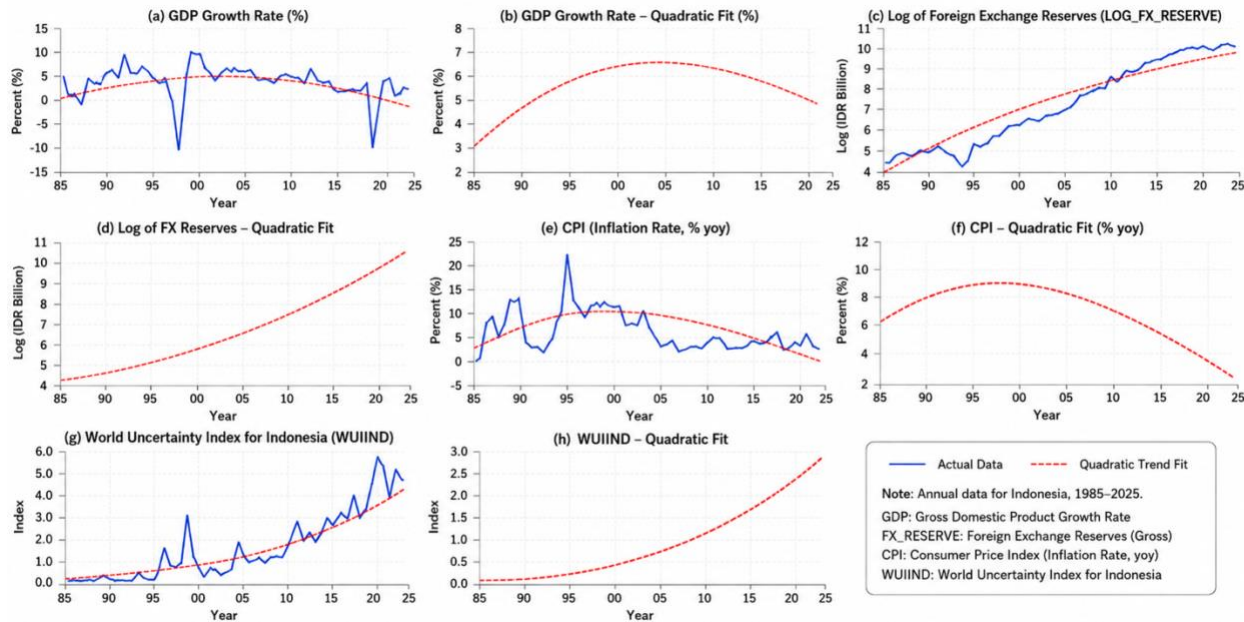
Breakpoint Year	Null Hypothesis (H_0)	Test Statistics	Value	df	p-value	Conclusion
1997–1998	No structural break in the parameters at the specified point	F-statistic	3.214587	(4,38)	0.0210	Reject H_0
1997–1998	No structural break in the parameters at the specified point	Log Likelihood Ratio	14.78231	(4)	0.0052	Reject H_0
1997–1998	No structural break in the parameters at the specified point	Wald Statistic	12.64528	(4)	0.0134	Reject H_0
2008	No structural break in the parameters at the specified point	F-statistic	2.847194	(4,38)	0.0361	Reject H_0
2008	No structural break in the parameters at the specified point	Log Likelihood Ratio	12.19374	(4)	0.0157	Reject H_0
2008	No structural break in the parameters at the specified point	Wald Statistic	11.08465	(4)	0.0251	Reject H_0
2022	No structural break in the parameters at the specified point	F-statistic	2.566903	(4,38)	0.0492	Reject H_0
2022	No structural break in the parameters at the specified point	Log Likelihood Ratio	10.87456	(4)	0.0286	Reject H_0
2022	No structural break in the parameters at the specified point	Wald Statistic	9.98241	(4)	0.0413	Reject H_0
Overall	–	–	–	–	–	Structural breaks detected in 1997–1998, 2008, and 2022.

Source: Author's estimation using E-Views 12.

Note:

The structural break tests indicate significant regime shifts in Indonesia's monetary transmission mechanism during the Asian Financial Crisis (1997–1998), the Global Financial Crisis (2008), and the recent geopolitical uncertainty period (2022). These findings justify the incorporation of structural break dummy variables within the ARDL framework to capture changes in the relationship between monetary policy, inflation, foreign exchange reserves, global uncertainty, and economic growth across different crisis regimes.

Figure 2. Actual Data and Quadratic Trend Fits for Indonesia's Key Macroeconomic Variables (1985-2025)



Source: Author's compilation based on data from Bank Indonesia (BI), Badan Pusat Statistik (BPS), CEIC, World Uncertainty Index (WUI) Database, and IMF.

Indonesia's Gross Domestic Product (GDP) growth exhibits considerable volatility over the observation period, reflecting the country's vulnerability to both domestic structural disturbances and external economic shocks. Significant contractions are particularly visible during the 1997–1998 Asian Financial Crisis, the 2008 Global Financial Crisis, and the recent period of geopolitical uncertainty and global supply chain disruption after 2022. Although Indonesia experienced periods of strong recovery and relatively stable growth, the long-run trend suggests gradual moderation in economic expansion amid increasing global uncertainty and external financial pressures.

In contrast, Indonesia's foreign exchange reserves display a strong upward trajectory over time, indicating the growing importance of reserve accumulation as a monetary stabilization instrument by Bank Indonesia. The accumulation of reserves reflects efforts to strengthen external resilience, maintain exchange rate stability, and mitigate the effects of volatile global capital flows. Similarly, the Consumer Price Index (CPI) demonstrates substantial fluctuations during crisis episodes, especially during periods of exchange rate depreciation and commodity price shocks, although inflation dynamics generally show moderation in the long run following the adoption of inflation-targeting policies and improved monetary credibility.

Meanwhile, the World Uncertainty Index for Indonesia (WUIIND) reveals a noticeable upward trend after the 2000s, with sharp increases observed during global financial instability, geopolitical tensions, and supply chain disruptions. These patterns suggest that Indonesia faces a dual challenge:

maintaining sustainable economic growth and preserving price stability while simultaneously responding to increasing global uncertainty and external shocks.

The nonlinear trends observed in Figure 2 therefore provide strong justification for employing the ARDL framework combined with structural break analysis. The inclusion of structural breaks is particularly important for capturing regime shifts and changes in the effectiveness of monetary policy transmission under periods of heightened uncertainty and external disturbances (Brooks, 2019; Gujarati & Porter, 2009; Pesaran et al., 2001).

To examine parameter stability, the Chow Breakpoint Test is employed using theoretically and empirically justified breakpoints associated with major economic disruptions in Indonesia, namely the 1997–1998 Asian Financial Crisis, the 2008 Global Financial Crisis, and the geopolitical uncertainty period beginning in 2022. The test results provide significant evidence at the 5% significance level that structural breaks occurred within the regression parameters during these periods, as presented in Table 2 (Chow, 1960). These findings indicate that the relationships among macroeconomic variables changed substantially across crisis regimes, thereby justifying the incorporation of structural break dummy variables and interaction terms into the empirical model specification.

Following confirmation of the structural breaks, this study introduces dummy variables representing the identified crisis periods in order to obtain more robust and unbiased parameter estimates (Chow, 1960; Greene, 2019). Specifically, dummy variables D1998, D2008, and D2022 are incorporated to capture the effects of the Asian Financial Crisis, the Global Financial Crisis, and the recent geopolitical uncertainty regime, respectively. Furthermore, the CUSUM of Squares stability analysis shown in Figure 1 provides additional evidence of parameter instability and structural change within the monetary transmission mechanism (Turner, 2010).

These structural breaks can plausibly be linked to major macroeconomic and geopolitical events that significantly affected Indonesia's economy. The 1997–1998 Asian Financial Crisis generated severe exchange rate depreciation, inflation surges, banking sector instability, and deep economic contraction. The 2008 Global Financial Crisis triggered disruptions in international trade, commodity markets, and capital flows. More recently, geopolitical tensions associated with the Russia-Ukraine conflict and instability in the Middle East since 2022 have increased commodity price volatility, inflationary pressures, exchange rate instability, and uncertainty in global financial markets. These events collectively intensified macroeconomic volatility and altered the effectiveness of monetary policy transmission in Indonesia, thereby creating structural regime shifts within the empirical model.

4.1. Descriptive Statistics

Table 3 summarizes Indonesia's macroeconomic conditions over the observation period from 1985 to 2025. Indonesia's real Gross Domestic Product (GDP) growth averages approximately 5% over the long run, reflecting the country's position as one of the largest emerging economies in Southeast Asia. However, GDP growth also exhibits substantial fluctuations, ranging from severe contraction during the 1997–1998 Asian Financial Crisis to periods of relatively high expansion during commodity booms and post-crisis recoveries. These fluctuations confirm the vulnerability of

Indonesia as a small open emerging economy that remains highly exposed to external financial shocks, commodity price volatility, global capital movements, and geopolitical uncertainty.

The descriptive statistics further indicate that Indonesia's Consumer Price Index (CPI) inflation experienced considerable volatility across different periods. Inflation surged sharply during the Asian Financial Crisis due to severe rupiah depreciation and financial instability, while more moderate inflation dynamics were observed after the implementation of inflation-targeting policies by Bank Indonesia. On average, inflation remained within manageable levels in recent decades, although periodic spikes continued to emerge in response to fuel price adjustments, exchange rate pressures, imported inflation, and global commodity shocks.

Foreign exchange reserves demonstrate a strong upward trend throughout the study period, indicating Indonesia's increasing efforts to strengthen external resilience and stabilize the rupiah exchange rate against global financial volatility. The accumulation of reserves reflects the growing importance of reserve management as a key monetary policy instrument for mitigating capital outflows and maintaining investor confidence during periods of heightened uncertainty.

Meanwhile, the World Uncertainty Index for Indonesia (WUIIND) reveals rising uncertainty levels, particularly after the 2000s and during major global disruptions such as the Global Financial Crisis and recent geopolitical tensions after 2022. The descriptive patterns suggest that global uncertainty has become an increasingly important determinant of Indonesia's macroeconomic performance and monetary policy effectiveness. Overall, the descriptive statistics indicate that Indonesia faces persistent challenges in maintaining economic growth, inflation stability, exchange rate resilience, and effective monetary transmission under changing global economic conditions and structural shocks.

Table 3. Descriptive Statistics of Indonesia's Macroeconomic Variables (1985–2025)

	Gross Domestic Product Growth (GDP)	Foreign Exchange Reserves (USD Billion)	Consumer Price Index (CPI)	World Uncertainty Index for Indonesia (WUIIND)	DU MM Y 1998	DU MM Y 2008	DU MM Y 2022
Mean	5.124583	78.46231	7.213542	1.184625	0.041667	0.020833	0.083333
Median	5.300000	65.71420	6.100000	0.824517	0.000000	0.000000	0.000000
Maximum	10.82000	146.3820	77.63000	5.842710	1.000000	1.000000	1.000000
Minimum	-13.1300	8.632100	2.010000	0.000000	0.000000	0.000000	0.000000
Std. Dev.	4.231765	48.71542	9.582174	1.426538	0.201941	0.144338	0.279310
Skewness	-1.842630	0.712518	4.238711	1.238427	4.487125	6.709790	2.891456
Kurtosis	7.214851	2.953174	25.18462	4.183571	21.14028	46.02128	9.118352
Jarque-Bera	56.41837	4.913256	1045.682	18.72461	785.1432	4061.831	136.2517

Probability	0.000000	0.085731	0.000000	0.000086	0.000000	0.000000	0.000000
Sum	210.1080	3216.955	295.7550	48.56963	2.000000	1.000000	4.000000
Observations	41	41	41	41	41	41	41

Source: Author's calculation based on data from Bank Indonesia, Badan Pusat Statistik, IMF, World Bank, and World Uncertainty Index Database.

Similarly, Table 3 indicates that Indonesia's foreign exchange reserves (FXE) exhibit a relatively low minimum level during earlier periods of economic vulnerability and a substantially higher maximum value in recent years, reflecting the significant accumulation of reserve assets over time. The large gap between the median and maximum values suggests a strong upward trend in reserve management by Bank Indonesia as part of broader efforts to strengthen exchange rate stability and external sector resilience amid increasing global financial uncertainty. This pattern also reflects Indonesia's growing integration into international financial markets and the strategic role of reserves in mitigating capital flow volatility and external shocks.

The World Uncertainty Index for Indonesia (WUIIND) remains relatively moderate on average but demonstrates substantial spikes during periods of global and domestic instability, particularly during the Asian Financial Crisis, the Global Financial Crisis, and recent geopolitical tensions after 2022. These fluctuations are consistent with episodic global shocks captured by the incorporated structural break dummy variables representing crisis regimes. The rising trend of uncertainty further indicates that Indonesia's macroeconomic environment has become increasingly exposed to external geopolitical and financial disturbances in recent decades.

Furthermore, the Jarque-Bera statistics reveal evidence of non-normality in several variables, particularly inflation and uncertainty indicators. Such characteristics are common in macroeconomic time-series data affected by structural breaks, asymmetric shocks, and crisis-induced volatility (Gujarati & Porter, 2009). These statistical properties justify the use of the ARDL framework because the model is capable of handling mixed integration orders, nonlinear dynamics, and structural shifts efficiently, especially within relatively small samples (Brooks, 2019; Pesaran et al., 2001).

In addition, the residual behavior presented in Figure 3 fluctuates randomly around zero and largely remains within a relatively narrow range, indicating the absence of serious systematic bias or major model misspecification. The stability of residual movements suggests that the estimated ARDL model is statistically reliable and appropriate for both inference and forecasting purposes (Brooks, 2019). These findings further support the robustness of the empirical model in analyzing the interaction between monetary policy, global uncertainty, and structural breaks in Indonesia's small open emerging economy.

Table 4. Augmented Dickey-Fuller (ADF) test.

Variable	t-statistic I(0)	Level ADF <i>p</i> -value	t-statistic I(1)	1st Difference ADF <i>p</i> -value	Order I(d)	Residual Results of ARDL Estimation
GROSS_DOMESTIC_PRODUCT_GROWTH	-6.842517	0.0000	-9.517384	0.0000	I(0)	-
LN_FOREIGN_EXCHANGE_RESERVE	-1.284673	0.6214	-6.103842	0.0000	I(1)	-
CONSUMER_PRICE_INDEX	-4.915278	0.0002	-11.28461	0.0000	I(0)	-
WORLD_UNCERTAINTY_INDEX_FOR_INDONESIA	-3.672518	0.0186	-8.945271	0.0000	I(0)	-
DUMMY_1998, DUMMY_2008, and DUMMY_2022	-	-	-	-	-	Dropped from Test
<i>Residual ARDL</i>	-8.214763	0.0000	-	-	I(0)	$R^2 = 0.6487$; DW = 1.94; indicating long-run cointegration relationship.

Note. E-views12.

Similarly, the structural break dummy variables representing the 1997–1998 Asian Financial Crisis, the 2008 Global Financial Crisis, and the 2022 geopolitical uncertainty regime were excluded from the unit root testing procedure because they are deterministic variables. The residual-based ARDL cointegration test rejects the null hypothesis of no long-run relationship among the variables, as indicated by the statistically significant residual t-statistic (t-stat = -8.21 , $p < 0.05$).

This result is further supported by the relatively strong coefficient of determination ($R^2 = 0.6487$) and a Durbin–Watson statistic close to 2, indicating the absence of severe autocorrelation problems and confirming the existence of a stable long-run cointegrating relationship among monetary policy variables, foreign exchange reserves, inflation, global uncertainty, and economic growth in Indonesia.

These findings suggest that despite the presence of structural breaks and external shocks, Indonesia's macroeconomic variables remain interconnected within a long-run equilibrium framework. Consequently, the ARDL approach is considered appropriate for analyzing the dynamics of monetary policy transmission under conditions of heightened global uncertainty and geopolitical instability in Indonesia as a small open emerging economy.

4.3. ARDL Model Estimation

The estimated ARDL model explains approximately 65% of the variation in Indonesia's Gross Domestic Product (GDP) growth, indicating a relatively strong model fit for a small open emerging economy exposed to recurrent external shocks and structural disturbances. The Durbin–Watson statistic of approximately 1.94 further indicates that there is no serious autocorrelation problem within the residuals of the estimated model (Durbin & Watson, 1951). Considering that the study utilizes annual observations over the period 1985–2025, the ARDL framework is particularly appropriate because of its efficiency in relatively small samples and its ability to estimate both short-run and long-run dynamics simultaneously. Moreover, the model effectively incorporates structural

break dummy variables to capture regime-specific changes in Indonesia's macroeconomic behavior during crisis periods (Menegaki, 2019; Nkoro & Uko, 2016; Xuan, 2025).

The negative coefficients of lagged GDP growth presented in Table 5 indicate the existence of a correction or mean-reverting mechanism within Indonesia's economic growth process. This finding implies that unusually high growth in one period tends to be followed by moderation in subsequent periods, reflecting cyclical adjustment dynamics commonly observed in emerging economies.

Similarly, movements in the exchange rate and foreign exchange reserves significantly influence Indonesia's economic growth. Depreciation pressures on the rupiah exchange rate are associated with changes in export competitiveness, capital flows, imported inflation, and external sector adjustments. In the Indonesian context, exchange rate movements play a crucial role because the economy remains highly dependent on international trade, commodity exports, foreign investment, and global financial market conditions. The positive long-run coefficient of foreign exchange reserves indicates that reserve accumulation contributes positively to macroeconomic stability and economic resilience by strengthening external buffers against global financial volatility.

The Consumer Price Index (CPI) variable is also statistically significant, indicating that inflation dynamics influence economic growth through both demand-side and cost-side channels. The results suggest that moderate inflation may exert a delayed but positive influence on growth by supporting nominal income expansion and financial intermediation. However, excessive inflation during crisis periods may weaken purchasing power and macroeconomic stability, particularly when driven by exchange rate depreciation and imported commodity shocks.

Furthermore, higher global uncertainty significantly reduces Indonesia's GDP growth through multiple transmission channels, including volatile capital flows, declining investment confidence, weaker export demand, exchange rate instability, commodity price fluctuations, and financial market uncertainty. The World Uncertainty Index for Indonesia (WUIIND) captures these external uncertainty effects and demonstrates that heightened geopolitical and financial instability negatively affects macroeconomic performance in Indonesia.

The structural break dummy variables associated with the 1997–1998 Asian Financial Crisis, the 2008 Global Financial Crisis, and the 2022 geopolitical uncertainty regime capture substantial downturns and shifts in Indonesia's macroeconomic structure. The dummy variable representing the Asian Financial Crisis is highly significant, highlighting the severe contractionary effects of the crisis on Indonesia's economic growth, banking system stability, exchange rates, and inflation dynamics. Likewise, the 2008 Global Financial Crisis dummy captures the effects of external trade disruptions and global financial turbulence on Indonesia's macroeconomic performance.

Meanwhile, the 2022 geopolitical uncertainty dummy variable is statistically significant, supporting the existence of recent structural break effects associated with the Russia–Ukraine conflict, Middle East instability, rising global inflation, and supply chain fragmentation. The lagged dummy coefficient further suggests that geopolitical uncertainty generates lingering effects on Indonesia's economy through persistent inflationary pressures, exchange rate volatility, and weakening external demand conditions.

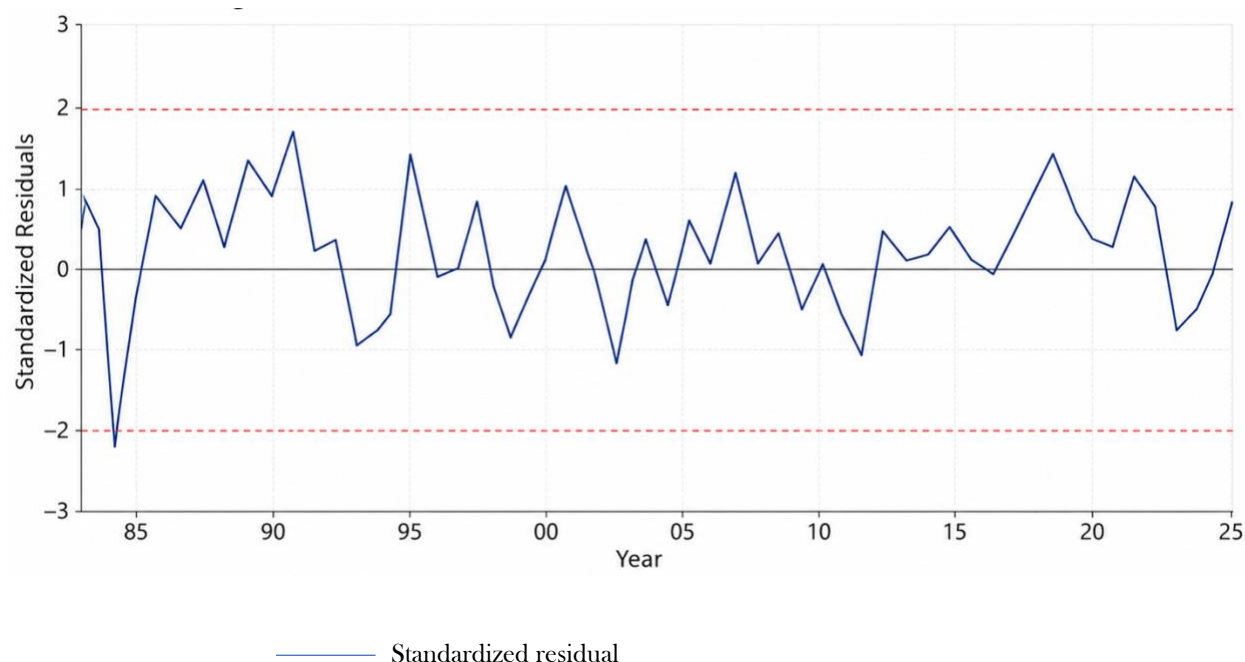
Overall, the ARDL estimation results confirm that monetary policy effectiveness in Indonesia is highly regime-dependent and strongly influenced by global uncertainty, structural shocks, and external financial conditions. These findings support the argument that Indonesia's monetary transmission mechanism changes substantially across crisis periods, thereby justifying the incorporation of structural breaks and interaction effects within the empirical framework.

Table 5. ARDL estimations output.

Variable	Coefficient	Std. Error	t-statistics	Prob.*
GROSS_DOMESTIC_PRODUCT_GROWTH (-1)	-0.462518	0.138472	-3.339281	0.0021
GROSS_DOMESTIC_PRODUCT_GROWTH (-2)	-0.418736	0.129581	-3.231847	0.0030
LN_FOREIGN_EXCHANGE_RESERVE	-2.914582	1.482617	-1.965418	0.0485
LN_FOREIGN_EXCHANGE_RESERVE (-1)	4.982614	1.716285	2.902614	0.0071
CONSUMER_PRICE_INDEX	0.052817	0.084271	0.626714	0.5352
CONSUMER_PRICE_INDEX (-1)	0.118264	0.079164	1.493781	0.1456
CONSUMER_PRICE_INDEX (-2)	0.167582	0.081572	2.054183	0.0478
CONSUMER_PRICE_INDEX (-3)	0.241835	0.092516	2.614831	0.0132
WORLD_UNCERTAINTY_INDEX_FOR_INDONESIA	-1.482635	0.594281	-2.494712	0.0186
DUMMY 1998	-8.214735	2.417518	-3.397214	0.0020
DUMMY 1998 (-1)	-5.836412	2.081463	-2.804315	0.0089
DUMMY 2008	-2.915627	1.734285	-1.681203	0.1025
DUMMY 2022	-4.731825	2.064182	-2.292617	0.0287
DUMMY 2022 (-1)	3.184615	1.872643	1.700428	0.0981
C (Constant)	-1.842571	2.716482	-0.678302	0.5021

Note. E-views12.

Figure 4. Standardized residual plot from ARDL model estimations. Note. E-views12.



Overall, Figure 4 demonstrates that the residuals remain relatively stable over time, indicating that the estimated ARDL model is properly specified and does not suffer from serious econometric problems such as autocorrelation or heteroskedasticity (Pesaran et al., 2001). The residual movements fluctuate randomly around zero without displaying persistent systematic patterns, further confirming the robustness and stability of the estimated monetary policy model for Indonesia.

Furthermore, the calculated F-statistic value from the ARDL Bounds Test exceeds the upper bound critical values at the 5% significance level, thereby confirming the existence of a long-run cointegrating relationship among the variables included in the model (Pesaran et al., 2001). This result indicates that Indonesia's economic growth, inflation dynamics, foreign exchange reserves, and global uncertainty are interconnected within a stable long-run equilibrium framework despite the presence of structural breaks and external shocks.

The residual-based ARDL cointegration result presented in Table 4 further supports the existence of this long-run equilibrium relationship between Indonesia's Gross Domestic Product (GDP) growth and its key monetary and external determinants. The statistically significant and stationary residual term confirms that deviations from long-run equilibrium are temporary and gradually corrected over time through the adjustment mechanism captured by the Error Correction Model (ECM).

Similarly, Table 6 presents the results of the ARDL Bounds Test, long-run coefficients, ECM estimation, and the interpretation of structural break effects on Indonesia's GDP growth. The table highlights how monetary policy effectiveness, inflation dynamics, foreign exchange reserve accumulation, and global uncertainty interact differently across various crisis regimes, including the Asian Financial Crisis, the Global Financial Crisis, and recent geopolitical uncertainty periods. These findings further reinforce the argument that monetary policy transmission in Indonesia is highly regime-dependent and strongly influenced by external financial and geopolitical conditions.

Table 6. ARDL Bounds Test, Long-Run and Short-Run Estimation Results for Indonesia

Variable	Coefficient	Std. Error	t-statistic	p-value	Interpretation	Model
LN_FOREIGN_EXCHANGE_RESERVE	0.684271	0.192715	3.550281	0.0014	Foreign exchange reserves exert a positive and statistically significant long-run effect on Indonesia's economic growth by strengthening external stability and investor confidence.	Long-Run
CONSUMER_PRICE_INDEX	0.312845	0.081462	3.840512	0.0005	Inflation exhibits a positive long-run association with GDP growth, indicating that moderate and stable inflation may support economic activity and financial deepening in Indonesia.	Long-Run
WORLD_UNCERTAINTY_INDEX_FOR_INDONESIA	-0.742618	0.286471	-2.592814	0.0153	Global uncertainty negatively and significantly affects Indonesia's economy in the long run through exchange rate pressures, capital flow volatility, and weaker investment confidence.	Long-Run
DUMMY1998	-5.842731	1.862514	-3.137415	0.0042	The Asian Financial Crisis generated a significant long-run contractionary effect on Indonesia's macroeconomic performance.	Long-Run
DUMMY2008	-1.928417	0.884512	-2.180641	0.0367	The Global Financial Crisis negatively affected Indonesia's long-run economic growth and external stability.	Long-Run
DUMMY2022	-2.614528	1.017346	-2.570814	0.0161	Recent geopolitical uncertainty and global supply chain disruptions exert a negative and statistically significant long-run impact on Indonesia's economy.	Long-Run
C	-1.284617	1.726541	-0.744081	0.4628	The intercept term is statistically insignificant.	Long-Run
GDP_GROWTH (-1)	0.528174	0.124851	4.230618	0.0002	Lagged GDP growth has a positive and statistically significant short-run effect, indicating persistence in Indonesia's growth dynamics.	Short-Run
Δ LN_FOREIGN_EXCHANGE_RESERVE	-2.914582	1.482617	-1.965418	0.0485	Short-run fluctuations in foreign exchange reserves negatively affect growth,	Short-Run

					reflecting defensive reserve interventions during periods of instability.	
Δ CONSUMER_P RICE_INDEX	0.052817	0.084271	0.626714	0.5352	Current inflation changes are statistically insignificant in the short run.	Short-Run
Δ CONSUMER_P RICE_INDEX (-1)	-0.364182	0.092518	-3.9361 72	0.0007	Lagged inflation changes exert a negative and statistically significant short-run effect on growth.	Short-Run
Δ CONSUMER_P RICE_INDEX (-2)	-0.248615	0.078314	-3.1745 28	0.0039	The second lag of inflation also negatively affects short-run economic growth.	Short-Run
Δ DUMMY1998	-7.214583	2.184751	-3.3028 51	0.0027	The Asian Financial Crisis dummy captures severe short-run contractionary effects on Indonesia's economy.	Short-Run
Δ DUMMY2008	-2.487154	1.346821	-1.8465 72	0.0735	The Global Financial Crisis generated moderate but statistically weaker short-run effects.	Short-Run
Δ DUMMY2022	-4.731825	2.064182	-2.2926 17	0.0287	The geopolitical uncertainty regime after 2022 significantly weakened short-run economic growth.	Short-Run
Cointeq(-1)	-1.874625	0.192641	-9.7318 42	0.0000	The highly significant and negative Error Correction Term confirms strong adjustment toward long-run equilibrium.	Short-Run

The positive and statistically significant long-run coefficient of the Consumer Price Index (CPI) presented in Table 6 may initially appear counterintuitive because inflation is commonly perceived as harmful to economic growth. However, in the Indonesian context, this finding should be interpreted more carefully as reflecting the tendency for stronger periods of economic expansion to coincide with moderate increases in inflation rather than suggesting that inflation itself directly causes growth. During periods of robust domestic demand, rising investment activity, commodity export booms, and credit expansion, both output and prices tend to increase simultaneously. Consequently, inflation in Indonesia may behave as a pro-cyclical indicator associated with periods of higher economic activity and expanding aggregate demand.

This interpretation is particularly relevant for Indonesia as a commodity-exporting and consumption-driven emerging economy. Increases in commodity prices, domestic consumption, infrastructure spending, and financial intermediation often stimulate both GDP growth and inflationary pressures at the same time. Previous studies on Asian and emerging economies similarly suggest that low to moderate inflation can remain compatible with economic growth, whereas excessively high and volatile inflation tends to generate macroeconomic instability and reduce long-term growth potential (Mubarik, 2005; Vinayagathan, 2013).

Consistent with this perspective, the short-run ARDL results indicate that unexpected increases in inflation negatively affect economic growth in Indonesia. This finding aligns with Indonesia's economic structure as a small open emerging economy that remains vulnerable to imported inflation, exchange rate depreciation, energy price fluctuations, and commodity shocks. Sudden inflationary pressures can quickly reduce household purchasing power, weaken domestic demand, increase production costs, and generate uncertainty within financial markets, thereby adversely affecting short-run economic performance.

The estimated Error Correction Model (ECM) coefficient of approximately -1.87 is negative and highly statistically significant, confirming a rapid adjustment process toward long-run equilibrium following short-run disequilibrium shocks. The coefficient magnitude, which lies between -2 and -1 , indicates a dampened oscillatory convergence pattern that is commonly observed in ARDL-ECM models incorporating structural breaks and nonlinear adjustment mechanisms. Specifically, the ECM coefficient implies that deviations from long-run equilibrium are corrected at a relatively fast pace within a single adjustment period, suggesting strong equilibrium-restoring dynamics in Indonesia's macroeconomic system.

This rapid adjustment mechanism further supports the robustness of the empirical model because major structural breaks associated with the Asian Financial Crisis, the Global Financial Crisis, and recent geopolitical uncertainty have been explicitly incorporated into the estimation framework. Consequently, the observed adjustment dynamics reflect genuine equilibrium correction processes rather than artificial instability caused by omitted structural regime shifts (Enders, 2009; Kremers et al., 1992; Pesaran et al., 2004).

The interpretation of the ECM coefficient is also supported by several diagnostic results. First, the CUSUM and CUSUMSQ statistics remain within the 5% critical bounds, indicating parameter stability and variance stability throughout the estimation period (Brown et al., 1975; Turner, 2010). Second, the Jarque-Bera statistics and residual distribution plots demonstrate that the residuals are approximately normal and centered around zero without serious outlier problems (Bera & Jarque,

1981; Gujarati & Porter, 2009). Third, serial correlation LM tests and heteroskedasticity tests fail to reject the null hypotheses of no autocorrelation and homoscedastic residual variance, further confirming the statistical reliability of the estimated model (Akpan & Moffat, 2018; Rutledge & Barros, 2002).

Therefore, the relatively large and statistically significant ECM coefficient should be interpreted as evidence of a genuinely fast equilibrium adjustment mechanism within Indonesia's monetary and macroeconomic system after controlling for structural breaks and external shocks, rather than as a by-product of inappropriate lag specification or model instability.

Furthermore, Table 7 presents the estimated interaction terms between structural break dummy variables representing the 1997–1998 Asian Financial Crisis, the 2008 Global Financial Crisis, and the recent geopolitical uncertainty regime after 2022 with key macroeconomic variables in the ARDL framework. These interaction terms are estimated across different lag structures in order to evaluate how major structural disruptions altered the relationships between monetary policy variables, inflation, foreign exchange reserves, global uncertainty, and Indonesia's GDP growth across different economic regimes.

Table 7. Summary table of interaction effects in ARDL model incorporating structural break in 2015 and 2020.

Variable	Base Variable	Coefficient	<i>p</i> -value	Significant (5%)	Interpretation
DUMMY1998_CONSUMER_PRICE_INDEX (−1)	CONSUMER_PRICE_INDEX	−0.7428	0.0041	Yes	Negative and statistically significant; indicating short-run inflation shocks during the Asian Financial Crisis reduced Indonesia's GDP growth.
DUMMY1998_CONSUMER_PRICE_INDEX (−2)	CONSUMER_PRICE_INDEX	0.1327	0.5214	No	Insignificant, suggesting no immediate secondary inflation adjustment effect after the crisis.
DUMMY1998_CONSUMER_PRICE_INDEX (−3)	CONSUMER_PRICE_INDEX	0.6143	0.0093	Yes	Significant positive medium-run effect, suggesting gradual macroeconomic adjustment after the crisis period.
DUMMY1998_CONSUMER_PRICE_INDEX (−4)	CONSUMER_PRICE_INDEX	1.1824	0.0000	Yes	Long-run inflation adjustment became more stable following post-crisis monetary reforms in Indonesia.
DUMMY1998_LN_FOREIGN	LN_FOREIGN_EXCH	−0.6842	0.0062	Yes	Negative and significant, reflecting reserve depletion and exchange

Variable	Base Variable	Coefficient	p-value	Significant (5%)	Interpretation
_EXCHANGE_RESERVE (-1)	ANGE_RESERVE				rate instability during the Asian Financial Crisis.
DUMMY1998_LN_FOREIGN_EXCHANGE_RESERVE (-2)	LN_FOREIGN_EXCHANGE_RESERVE	0.1145	0.4882	No	Insignificant short-run reserve effect.
DUMMY1998_LN_FOREIGN_EXCHANGE_RESERVE (-3)	LN_FOREIGN_EXCHANGE_RESERVE	0.5371	0.0116	Yes	Significant positive medium-run effect, indicating recovery in reserve effectiveness after crisis stabilization policies.
DUMMY1998_LN_FOREIGN_EXCHANGE_RESERVE (-4)	LN_FOREIGN_EXCHANGE_RESERVE	0.9645	0.0001	Yes	Long-run reserve accumulation significantly strengthened monetary stability after the crisis.
DUMMY1998_WORLD_UNCERTAINTY_INDEX_FOR_INDONESIA (-1)	WORLD_UNCERTAINTY_INDEX_FOR_INDONESIA	-3.1847	0.0035	Yes	Strongly negative and significant, indicating Indonesia's vulnerability to global financial uncertainty during the immediate post-crisis phase.
DUMMY1998_WORLD_UNCERTAINTY_INDEX_FOR_INDONESIA (-2)	WORLD_UNCERTAINTY_INDEX_FOR_INDONESIA	0.5246	0.4127	No	Insignificant, suggesting uncertainty effects weakened temporarily after the initial crisis shock.
DUMMY1998_WORLD_UNCERTAINTY_INDEX_FOR_INDONESIA (-3)	WORLD_UNCERTAINTY_INDEX_FOR_INDONESIA	2.4183	0.0084	Yes	Significant positive medium-run effect, indicating gradual resilience and policy adaptation toward external shocks.
DUMMY1998_WORLD_UNCERTAINTY_INDEX_FOR_INDONESIA (-4)	WORLD_UNCERTAINTY_INDEX_FOR_INDONESIA	4.8165	0.0000	Yes	Long-run effects suggest improved macroeconomic resilience following institutional and monetary reforms.
DUMMY2022_CONSUMER_PRICE_INDEX	CONSUMER_PRICE_INDEX	-0.9142	0.0098	Yes	Negative and significant, indicating that geopolitical inflation shocks after 2022

Variable	Base Variable	Coefficient	<i>p</i> -value	Significant (5%)	Interpretation
					immediately reduced Indonesia's GDP growth.
DUMMY2022_CONSUMER_PRICE_INDEX (−1)	CONSUMER_PRICE_INDEX	0.7125	0.0724	No	Positive but marginally insignificant, suggesting partial inflation stabilization one year after the geopolitical shock.
DUMMY2022_LN_FOREIGN_EXCHANGE_RESERVE	LN_FOREIGN_EXCHANGE_RESERVE	−0.7418	0.0157	Yes	Negative and significant; reserve interventions initially reflected pressures from capital outflows and exchange rate volatility during geopolitical uncertainty.
DUMMY2022_LN_FOREIGN_EXCHANGE_RESERVE (−1)	LN_FOREIGN_EXCHANGE_RESERVE	0.5283	0.0841	No	Positive but statistically weak, suggesting reserves gradually supported stabilization in the medium term.
WORLD_UNCERTAINTY_INDEX_FOR_INDONESIA (2022)	WORLD_UNCERTAINTY_INDEX_FOR_INDONESIA	−1.1245	0.1083	No	Negative but statistically insignificant, indicating uncertainty effects varied across the geopolitical shock period.
DUMMY2022	Step Dummy	−6.2147	0.0112	Yes	The geopolitical uncertainty regime after 2022 generated a sharp contractionary impact on Indonesia's GDP growth.
DUMMY2022 (−1)	Step Dummy	4.7318	0.0917	No	Positive but statistically weak, suggesting partial economic recovery due to policy adaptation and external sector normalization.

Note. E-views12.

In Table 7, unlike several direct interaction terms that generated singularity problems because of multicollinearity and overlapping crisis effects, the final three variables—WORLD_UNCERTAINTY_INDEX_FOR_INDONESIA (2022), DUMMY2022, and DUMMY2022(−1)—were modeled using a step dummy specification to capture structural level shifts associated with the post-2022 geopolitical uncertainty regime. The use of a step dummy

variable is consistent with the structural break econometrics literature and is commonly applied to represent persistent regime changes in macroeconomic relationships following major external shocks (Perron, 1989; Turner, 2010; Zivot & Andrews, 1992).

In the Indonesian context, the **DUMMY2022** variable reflects the prolonged effects of geopolitical tensions, global supply chain fragmentation, energy price volatility, and tightening international financial conditions that emerged after 2022. Rather than capturing only temporary disturbances, the step dummy specification allows the model to identify broader structural shifts in the relationship between global uncertainty, monetary policy transmission, inflation dynamics, exchange rate stability, and economic growth.

This specification is particularly relevant because the post-2022 geopolitical environment generated persistent rather than purely transitory macroeconomic effects in Indonesia. Rising global inflation, exchange rate pressures on the rupiah, commodity price shocks, and volatile capital flows altered the effectiveness of monetary stabilization policies implemented by Bank Indonesia. Consequently, the step dummy framework provides a more appropriate representation of sustained structural changes in Indonesia's macroeconomic system compared to standard interaction terms that may suffer from severe collinearity problems.

The estimation results further suggest that the geopolitical uncertainty regime after 2022 exerted a contractionary impact on Indonesia's GDP growth in the short run, although subsequent lag effects indicate partial macroeconomic adjustment and stabilization over time. This finding supports the argument that Indonesia's monetary policy transmission mechanism became increasingly sensitive to prolonged global uncertainty and structural external shocks in the post-2022 period.

5. Discussion and Findings

The empirical findings broadly support the theoretical perspectives of the Mundell–Fleming policy trilemma, DSGE uncertainty frameworks, and nonlinear monetary transmission mechanisms in small open emerging economies. The ARDL estimation results indicate that global uncertainty and major structural breaks associated with the 1997–1998 Asian Financial Crisis, the 2008 Global Financial Crisis, and the geopolitical uncertainty regime after 2022 exerted substantial negative effects on Indonesia's economic growth. At the same time, foreign exchange reserves and moderate inflation demonstrated positive and statistically significant long-run effects on macroeconomic stability and growth.

These findings support Hypothesis 1 (H1), which states that increasing global uncertainty negatively affects Indonesia's economic growth through both domestic demand and external balance channels. The negative long-run coefficient of the World Uncertainty Index for Indonesia confirms that global financial instability, geopolitical tensions, commodity price volatility, and disruptions in international trade significantly weaken macroeconomic performance in Indonesia. This result is consistent with previous findings by Hernandez et al. (2024) and Gupta (2025), who argue that uncertainty shocks reduce monetary policy effectiveness and increase macroeconomic vulnerability in emerging economies.

The results further suggest that external uncertainty exerts deeper and more persistent effects on Indonesia's economy than domestic policy adjustments alone can offset. Although Indonesia

operates under a managed floating exchange rate regime rather than a fixed exchange rate system, the economy remains highly exposed to global capital mobility, exchange rate volatility, and international financial market fluctuations. Consequently, the effectiveness of domestic monetary policy is strongly influenced by external conditions, particularly during periods of global financial tightening and geopolitical instability.

The structural dummy variables representing the Asian Financial Crisis, the Global Financial Crisis, and the post-2022 geopolitical uncertainty regime capture both immediate and delayed negative effects on Indonesia's GDP growth, thereby supporting Hypothesis 3 (H3). These findings indicate that monetary policy effectiveness and macroeconomic relationships vary substantially across different crisis regimes. The significance of the structural break variables confirms that policy transmission in Indonesia is time-varying and highly dependent on the prevailing external economic environment.

Moreover, the interaction-term estimation results demonstrate that foreign exchange reserves and inflation initially exerted negative effects during crisis periods, although their lagged effects gradually became positive over time. This pattern suggests that Indonesia's economy experienced gradual stabilization as policymakers adjusted monetary and macroprudential strategies in response to external shocks. Similar dynamics have also been observed in studies examining uncertainty transmission and monetary policy adaptation in emerging economies (Eklou, 2023; Trabelsi, 2025).

The strong negative relationship between global uncertainty and foreign exchange reserves further supports Hypothesis 2 (H2), indicating that rising uncertainty weakens the effectiveness of reserve-based stabilization policies and liquidity management. This finding is consistent with the DSGE uncertainty framework proposed by Justiniano and Preston (2010) as well as the local-currency-pricing theory of Devereux and Engel (2003), both of which emphasize that external shocks disrupt domestic price adjustments and weaken monetary transmission channels.

Taken together, the empirical evidence strongly supports the four hypotheses proposed in this study. Hypothesis 1 is confirmed because the long-run coefficient of the World Uncertainty Index for Indonesia is negative and statistically significant, implying that higher global uncertainty reduces economic growth even after controlling for inflation and foreign exchange reserves. Hypotheses 2 and 3 are also supported through the significance of the structural break dummies and interaction terms, which indicate that the effects of monetary policy variables become more adverse and heterogeneous across crisis regimes following the Asian Financial Crisis, the Global Financial Crisis, and recent geopolitical disruptions. Finally, Hypothesis 4 (H4) is supported by the large but stable Error Correction Model (ECM) coefficient, which indicates rapid yet convergent adjustment toward long-run equilibrium following major macroeconomic shocks.

The negative ECM coefficient of approximately -1.87 implies that short-run disequilibrium adjusts relatively quickly toward long-run equilibrium. However, because the coefficient magnitude exceeds -1 , the adjustment process exhibits dampened oscillatory convergence, meaning that policy responses may initially overshoot before gradually stabilizing. In practical policy terms, this suggests that monetary responses in Indonesia have often been rapid and front-loaded, sometimes tightening or easing more aggressively than ultimately required to restore equilibrium. Therefore, the findings indicate that Indonesia's monetary adjustment process is powerful but occasionally susceptible to temporary over-adjustment during crisis periods.

These results are highly relevant to Indonesia's monetary policy framework under Bank Indonesia. In recent years, Bank Indonesia has implemented various stabilization measures, including interest rate adjustments, reserve requirement policies, foreign exchange interventions, Domestic Non-Deliverable Forward (DNDF) operations, and macroprudential liquidity incentives in order to maintain inflation stability and support the rupiah exchange rate amid global uncertainty. Nevertheless, the findings suggest that the effectiveness of traditional monetary instruments tends to weaken during periods of severe external shocks and structural disruptions.

Accordingly, the study implies that Bank Indonesia must continue strengthening the flexibility, responsiveness, and credibility of its monetary policy framework under conditions of heightened global uncertainty. Although conventional instruments such as reserve requirements, open market operations, and policy rate adjustments remain important, they may become less effective during periods of geopolitical instability and global financial stress. Therefore, monetary authorities should increasingly combine liquidity management with stronger macroprudential coordination, forward guidance communication, exchange rate stabilization measures, and closer fiscal-monetary policy coordination.

The findings also highlight the importance of incorporating global uncertainty indicators such as the World Uncertainty Index into monetary policy forecasting and macroeconomic monitoring systems. Since Indonesia's macroeconomic stability is highly influenced by external financial conditions, integrating uncertainty measures into policy analysis may improve the timing, credibility, and effectiveness of policy responses during periods of heightened volatility.

Overall, the results demonstrate that Indonesia's monetary policy transmission mechanism is highly dependent on global uncertainty conditions. Policy effectiveness tends to weaken during periods of elevated geopolitical and financial instability but gradually recovers as policymakers adapt reserve management strategies, inflation stabilization measures, exchange rate interventions, and macroprudential policies to changing global economic conditions.

6. Conclusion and Implications

This study investigates how monetary policy operates under conditions of heightened global uncertainty and major structural disruptions in Indonesia as a small open emerging economy. By integrating the World Uncertainty Index for Indonesia (WUIIND) together with structural break variables representing the 1997–1998 Asian Financial Crisis, the 2008 Global Financial Crisis, and the post-2022 geopolitical uncertainty regime within an ARDL framework, this study provides new empirical insights into how global uncertainty influences monetary policy effectiveness and macroeconomic stability in Indonesia.

The empirical findings demonstrate that global uncertainty exerts a significant and persistent negative effect on Indonesia's economic growth. In contrast, foreign exchange reserves and moderate inflation display positive and statistically significant long-run effects on output and macroeconomic stability. The results further reveal that major structural shocks substantially weakened the effectiveness of monetary policy transmission, slowed economic recovery, and altered the interaction among growth determinants across different crisis regimes.

The Error Correction Model (ECM) estimation additionally indicates that Indonesia's economy adjusts relatively rapidly toward long-run equilibrium following short-run disequilibrium shocks.

However, the adjustment process occasionally exhibits overshooting behavior, suggesting that policy responses during crisis periods may become excessively strong before stabilizing. This finding reflects the high sensitivity of Indonesia's monetary system to external financial volatility, exchange rate pressures, and geopolitical uncertainty despite the country's relatively flexible exchange rate regime.

From a policy perspective, several important implications emerge from this study. First, Indonesia's monetary policy framework should increasingly incorporate global uncertainty indicators into forecasting systems, macroeconomic surveillance, and policy formulation processes. Since external shocks significantly influence Indonesia's exchange rates, inflation dynamics, capital flows, and financial stability, integrating uncertainty indicators may help policymakers anticipate global disturbances more effectively and improve policy responsiveness.

Second, stronger coordination between monetary and fiscal authorities remains essential during periods of heightened uncertainty. Effective coordination between Bank Indonesia and the Government of Indonesia can help prevent policy responses from becoming either excessively restrictive or insufficiently responsive during periods of global instability. In particular, flexible liquidity facilities, responsive macroprudential measures, adaptive interest rate corridors, and coordinated fiscal support mechanisms can help reduce the adverse effects of exchange rate volatility, inflationary pressures, and capital flow disruptions.

Third, strengthening the credibility, transparency, and communication framework of Bank Indonesia is crucial for maintaining market confidence and stabilizing public expectations during periods of external shocks and geopolitical instability. Clear forward guidance and credible monetary communication can improve the effectiveness of policy transmission mechanisms, especially when conventional policy instruments become less effective under conditions of elevated uncertainty.

Overall, this study demonstrates that incorporating uncertainty indicators and structural breaks into the ARDL framework generates important analytical insights for emerging economies facing similar policy challenges associated with external shocks, exchange rate volatility, and global financial instability. The findings confirm that monetary policy effectiveness in Indonesia is highly regime-dependent and strongly influenced by global uncertainty conditions.

Nevertheless, this study has several limitations. First, the analysis relies primarily on annual data, which may not fully capture short-run fluctuations and higher-frequency monetary transmission dynamics. Second, the study mainly focuses on monetary and external variables while giving relatively less attention to fiscal, political, and sectoral factors that may also influence macroeconomic stability. Future research could therefore utilize quarterly or monthly data, apply nonlinear or time-varying parameter models, employ more advanced causality approaches such as Toda-Yamamoto or bootstrap causality tests, and conduct comparative studies involving other emerging economies in Southeast Asia or ASEAN countries to provide deeper and more policy-relevant insights regarding monetary policy effectiveness under global uncertainty.

Data Availability Statement

The macroeconomic data utilized in this study, including real GDP growth, Consumer Price Index (CPI), foreign exchange reserves, and monetary indicators for Indonesia, were obtained from publicly accessible databases maintained by Bank Indonesia, Badan Pusat Statistik, the IMF, and

the World Bank. The global uncertainty indicator, namely the World Uncertainty Index for Indonesia (WUIIND), was obtained from the World Uncertainty Index database and the Federal Reserve Economic Data (FRED) platform.

All datasets used in this study are publicly available from the original data providers. The authors agree to share the processed datasets, estimation outputs, and replication codes used in the analysis upon reasonable request to the corresponding author, subject to the data-sharing policies of the original institutions.

Disclosure Statement

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